

HTS-MultiResUNet: Dual-Attention Hybrid CNN-Transformer with Edge-Aware Loss for Multimodal Biomedical Image Segmentation

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Abstract—Medical image segmentation is an important prerequisite for accurate clinical diagnosis and treatment planning. Although MultiResUNet has mitigated certain structural limitations of the standard U-Net architecture through multi-scale feature extraction, it remains susceptible to false-positive activations and fragmented masks due to a lack of global contextual reasoning and semantic gaps along conventional skip connections. To address these limitations, we propose HTS-MultiResUNet, a hybrid CNN-Transformer architecture. The model integrates Squeeze-and-Excitation (SE) mechanisms into multi-scale extraction blocks for adaptive channel-wise feature recalibration and implements a Multi-Head Self-Attention (MHSA) bottleneck at the deepest layer to capture long-range global contextual dependencies. Furthermore, the conventional skip connections are modified into Attention-aware Residual Paths (Att – ResPath) to reduce low-level background noise prior to decoder fusion. Additionally, an Edge-Aware Focal Tversky Loss (EA – FTL) incorporating a discrete Sobel operator is introduced to handle class imbalances and enforce boundary precision. Evaluation using a five-fold cross-validation protocol across four distinct imaging modalities indicates that HTS-MultiResUNet improves segmentation performance compared to the baseline MultiResUNet. Specifically, the proposed model achieves relative increases in the Jaccard Index of 1.10% on Fluorescence Microscopy, 0.76% on Electron Microscopy, 5.83% on Dermoscopy, and 1.65% on Endoscopy, while reducing validation variance and providing better boundary delineation in the presence of imaging artifacts.

Index Terms—Biomedical image segmentation, MultiResUNet, Multi-Head Self-Attention, Squeeze-and-Excitation, Edge-Aware Loss, Att-ResPath, Multimodal imaging.

I. INTRODUCTION

MEDICAL Image Segmentation (MIS) is an important component in computer-aided diagnosis systems, enabling the identification of organs, tumors, lesions, and other anatomical structures from multimodal medical images [1], [2]. However, pixel-level segmentation remains challenging due to low-contrast boundaries, variations in object shapes, imaging noise, and inter-patient variability [3]. Traditional segmentation approaches relying on handcrafted features and clustering methods often exhibit limited robustness and generalization capability in complex medical imaging scenarios, occasionally failing to adapt to subtle clinical variations [4].

Recent developments in deep learning, particularly Convolutional Neural Networks (CNNs), have improved biomedical image segmentation performance [5], [6]. Among existing architectures, U-Net [7] has established a foundational framework owing to its symmetric encoder-decoder structure and skip connections that preserve low-level spatial information during feature reconstruction. Numerous variants of U-Net have subsequently been proposed and tailored for specific medical imaging applications.

Despite its widespread adoption, the standard U-Net architecture presents certain architectural limitations. Conventional convolution operations primarily capture local spatial features within a restricted receptive field, which limits their capacity to model long-range global contextual dependencies. In addition, the semantic discrepancy between encoder and decoder features can reduce the effectiveness of direct skip connections, potentially introducing low-level noise into high-level representations [8]. Furthermore, standard CNN-based models may struggle to learn target regions characterized by substantial class imbalance relative to the background, which can lead to biased predictions and sub-optimal boundary delineations. To address these issues, MultiResUNet [9] introduced MultiRes blocks and residual paths to enhance multi-scale feature extraction and mitigate semantic feature propagation mismatches. However, because MultiResUNet relies predominantly on localized convolutional operations, its global context modeling capacity remains constrained.

Recently, attention mechanisms and Transformer-based architectures have demonstrated the capacity to capture long-range dependencies in various computer vision tasks [10], [11]. Hybrid CNN-Transformer frameworks, such as TransUNet [12] and UNETR [13], have shown that global self-attention mechanisms can complement convolutional representations in medical image segmentation. Moreover, channel attention mechanisms, including Squeeze-and-Excitation (SE) networks [14], enhance feature discrimination by adaptively recalibrating informative feature channels and suppressing irrelevant noise.

However, the continuous integration of these methodologies reveals certain optimization challenges when deployed across distinct biomedical imaging modalities. First, conventional convolutional operations treat all feature channels uniformly, which can propagate redundant spatial details and background noise. In clinical benchmarks, this redundancy is often manifested as high-frequency artifacts, such as ruler

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markings, skin hair, and faded ink spots in dermoscopic images, or specular reflections and complex mucosal textures in endoscopic sequences. Without adaptive channel feature recalibration, pure multi-scale convolutional layers may be more prone to fragmented predictions. Second, while Vision Transformers are well-suited for global contextual reasoning, their quadratic computational complexity $\mathcal{O}(N^2)$ restricts their application on high-resolution, dense spatial representations, often inducing substantial memory overhead during the decoding phase. Lastly, standard region-based loss functions, such as Binary Cross-Entropy or standard Dice coefficients, optimize global pixel-wise overlaps but are intrinsically invariant to localized topological borders. Consequently, under conditions of substantial spatial class imbalance—such as micro-nuclear overlapping in fluorescence microscopy or structural ambiguity in transmission electron microscopy—networks optimized via conventional losses frequently yield poorly defined morphological boundaries and elevated false-positive regions.

To address these limitations, this paper details the Hybrid Transformer and Squeeze-Excitation MultiResUNet (HTS-MultiResUNet) architecture. The proposed framework integrates Transformer and attention mechanisms into the MultiResUNet framework to balance local feature extraction and global contextual reasoning. Architecturally, Squeeze-and-Excitation (SE) mechanisms are integrated into the MultiRes blocks and ResPath modules. This modification allows the network to recalibrate channel-wise features, emphasizing informative patterns while reducing the impact of background noise prior to feature fusion. To capture global context without significantly increasing computational overhead, a Hybrid Transformer Bottleneck incorporating Multi-Head Self-Attention (MHSA) is implemented at the deepest, lowest-resolution layer. This placement restricts self-attention calculations to a lower spatial resolution where spatial dimensions are minimized, modeling long-range dependencies within the compressed latent space.

Furthermore, to address boundary degradation and class imbalance, the network is optimized using an Edge-Aware Focal Tversky Loss (EA-FTL) function. This composite objective combines a region-based Focal Tversky component, which shifts gradient weights toward hard-to-classify pixels in imbalanced datasets, with a parallel edge-aware component utilizing a discrete derivative Sobel operator. By penalizing the Mean Absolute Error (MAE) between the predicted and reference gradient edge maps, the loss function enforces structural edge constraints and is designed to reduce false-positive responses outside the target regions, leading to sharper and more consistent segmentation boundaries. Evaluation across four distinct imaging modalities is conducted to analyze the performance, convergence characteristics, and stability of the architecture relative to standard baselines.

The primary contributions of this work are summarized as follows:

- We present HTS-MultiResUNet, a hybrid architecture combining multi-scale convolutions, channel attention, and Transformer-based self-attention for biomedical image segmentation.

- We modify the baseline MultiRes blocks and ResPath modules by integrating Squeeze-and-Excitation (SE) mechanisms to perform adaptive channel-wise feature recalibration and mitigate background noise propagation.
- We implement a localized Transformer bottleneck at the lowest-resolution layer to model long-range global contextual relationships, complementing the localized receptive fields of the convolutional layers.
- We employ an Edge-Aware Focal Tversky Loss (EA-FTL) function incorporating a Sobel-based boundary penalty to structurally optimize segmentation performance along complex boundaries and in regions with substantial class imbalance.
- We provide a comprehensive experimental evaluation using a five-fold cross-validation protocol across four distinct biomedical datasets, assessing the performance gains, parameter trade-offs, and stability of the proposed configuration compared to the standard MultiResUNet baseline.

The remainder of this paper is organized as follows. Section IV provides an overview of the baseline MultiResUNet architecture. Section III details the design motivations and theoretical considerations underlying our hybrid approach. Section IV elucidates the methodology, structural design, and edge-aware optimization strategy of the proposed HTS-MultiResUNet. Section V and Section VI present the dataset configurations and experimental methodologies, respectively. Section VII discusses the quantitative and qualitative results, and Section VIII concludes the paper.

II. RELATED WORK AND RESEARCH GAP ANALYSIS

To contextualize the architectural choices of HTS-MultiResUNet, this section reviews contemporary state-of-the-art (SOTA) biomedical image segmentation frameworks across five core design domains. Rather than providing a historical overview, we examine the structural limits and optimization constraints that motivate our hybrid model. A comparative summary of these structural attributes is organized in the taxonomy matrix presented in Table I.

A. Conventional U-Net and Multi-Scale CNN Variants

The classic U-Net framework [7] established the baseline encoder-decoder blueprint for medical imaging, utilizing a symmetric layout where the contracting path extracts contextual representations and the expanding path reconstructs pixel-wise mask alignments, as schematically illustrated in Fig. 1. The encoder follows a design where each of the four progressive downsampling stages consists of two successive 3×3 convolutional layers, each followed by a Rectified Linear Unit (ReLU) activation function and a 2×2 max pooling operation with a stride of 2. Along this contraction, spatial resolution is halved at each stage while the number of filters doubles to maintain representational capacity. At the deepest bottleneck layer, two sequential 3×3 convolutional layers bridge the pathways without pooling.

Symmetrically, the decoder pathway restores the spatial coordinates through 2×2 transposed convolutions, which halve

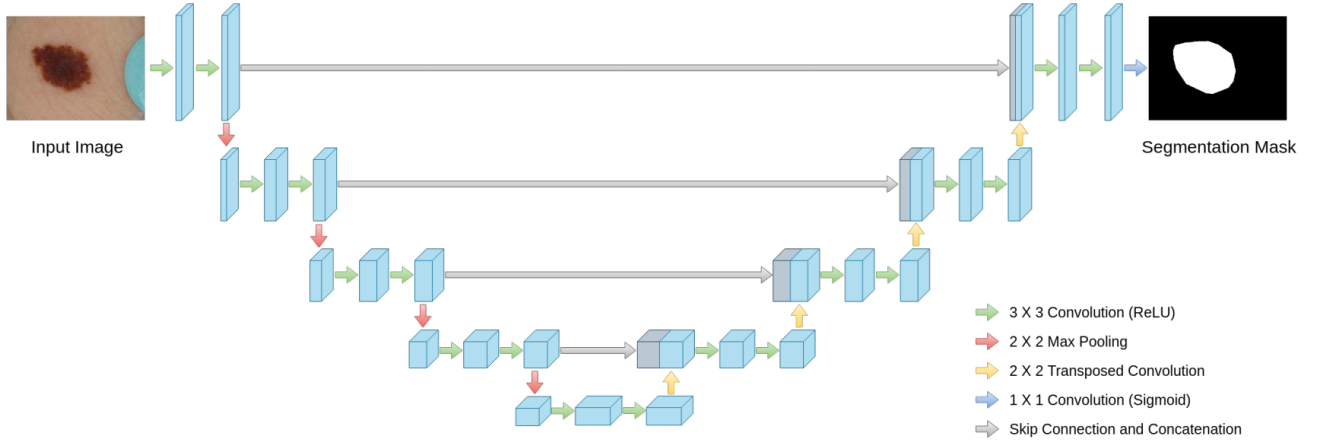


Fig. 1: The original U-Net architecture comprises an encoder pathway, a symmetric decoder pathway, and direct skip connections between corresponding resolution levels for feature concatenation.

the feature channels at each expanding level. Following each upsampling step, the feature streams are concatenated along the channel axis with the corresponding high-resolution maps from the contracting path via skip connections to preserve fine-grained edge profiles. This combined tensor is subsequently processed by two consecutive 3×3 convolution operations. This expanding process is repeated four times until the original input resolution is recovered. Finally, a 1×1 convolution layer maps the feature channels to the predicted binary mask using a sigmoid activation function.

To expand the single-scale receptive field constraint of standard U-Net, MultiResUNet [9] introduced the MultiRes block, detailed in Fig. 2. Instead of stacked uniform convolutions, it factorizes large 5×5 and 7×7 kernels into parallel, stacked sequences of hierarchical 3×3 convolutions to capture multi-scale properties adhering to spatial factorization principles [15].

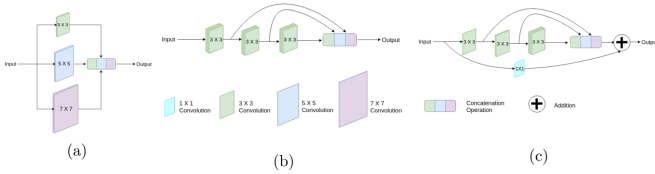


Fig. 2: The MultiRes block factorizes large 5×5 and 7×7 filters into a sequence of stacked 3×3 layers, incorporating a residual connection and progressive filter scaling for multi-scale feature representation.

The filter distribution is split across these parallel streams based on a base block width parameter W , computed as a scaled variant of the equivalent U-Net filter count according to Equation (1):

$$W = \alpha \times U \quad (1)$$

where α is a scaling coefficient calibrated to maintain parameter parity with standard configurations.

To address the semantic gap between pathways, the ResPath configuration in MultiResUNet [9] incorporates a sequence of non-linear convolutional layers, reducing structural discrepancies at deeper layers by utilizing four, three, two, and one convolutional blocks along the progressive pathways, respectively, as shown in the full baseline layout in Fig. 4.

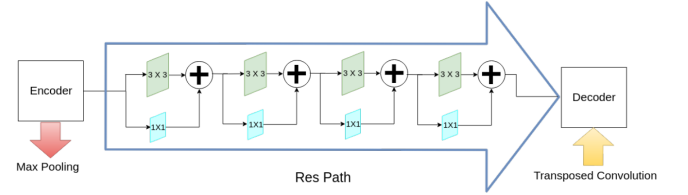


Fig. 3: The Res path replaces direct skip connections with a sequence of convolutional layers and residual connections to bridge the semantic gap between encoder and decoder features.

Research Gap: Despite these multi-scale advancements, both U-Net and MultiResUNet rely primarily on localized, isotropic convolution operations. Consequently, they process spatial zones uniformly, limiting their capacity to model long-range global contextual relationships. This structural constraint can lead to prediction fragmentation and false-positive mask bleeding when encountering complex organ topologies or highly artifact-laden clinical backgrounds.

B. Attention-Enhanced Segmentation Frameworks

To expand the spatial representations of conventional CNNs, recent frameworks integrated attention mechanisms to dynamically emphasize informative target regions. Attention U-Net [16] implemented spatial gating blocks along the skip paths, utilizing high-level contextual vectors from deeper layers to filter out low-level redundant background noise before concatenation. In parallel, channel attention methods, such as the Squeeze-and-Excitation (SE) Network [14], perform adaptive channel-wise feature recalibration by modeling inter-channel dependencies via global average pooling and multi-

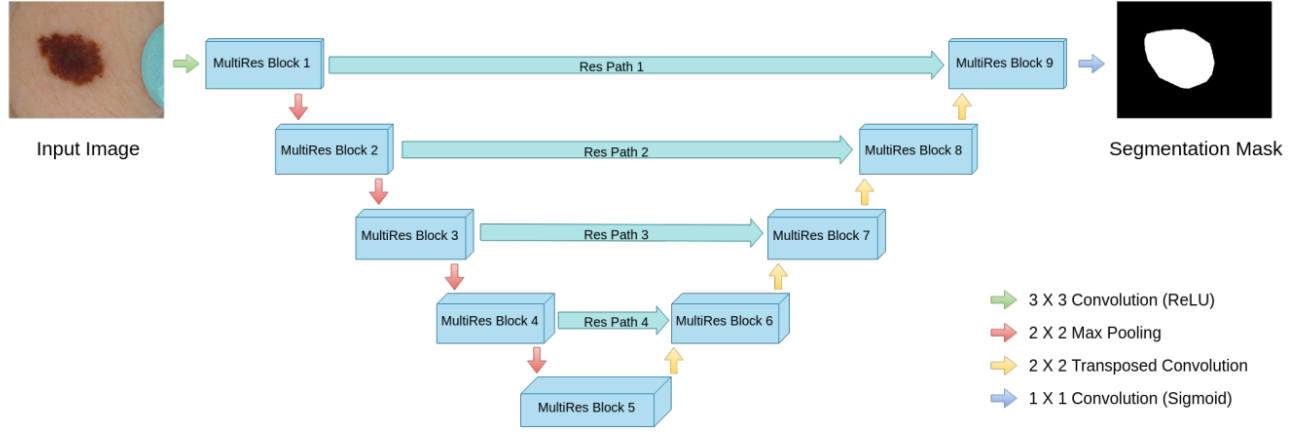


Fig. 4: Overall MultiRes U-Net Architecture proposed by Ibtehaz et al. [9]

layer perceptron blocks, suppressing non-informative feature maps.

Recent advancements have introduced deeper feature calibration schemes; for instance, the SDI (Semantics and Detail Infusion) module in contemporary U-Net configurations [17] selectively gates cross-path feature integration to prevent low-level noise propagation during decoder reconstruction.

Research Gap: Spatial and channel attention mechanisms remain inherently bounded by the local representations of the underlying convolutional feature maps. While they improve localized feature selection, they lack the global contextual reasoning required to establish topological continuity, often generating discontinuous contours when ambiguous tissue boundaries share highly overlapping intensity profiles with surrounding structures.

C. Transformer-Based Medical Image Segmentation

Leveraging sequence-to-sequence learning, Vision Transformers (ViTs) capture global relationships via Multi-Head Self-Attention (MHSA). TransUNet [12] deployed a tokenized ViT encoder backbone coupled with a convolutional decoder, capturing long-range spatial correlations across a grid sequence. Conversely, UNETR [13] introduced a pure 3D Transformer encoder, utilizing direct patch tokenization combined with deconvolutional skip highways to project multi-scale representations back to dense prediction spaces.

More recent hybrid paradigms, such as HiFormer [18] and HTC-Net [19], have attempted to bridge these domains by harmonizing local feature extraction with global contextual reasoning through hierarchical multi-scale fusion pathways. Furthermore, frameworks incorporating large-kernel pyramid convolutions combined with lightweight multi-layer perceptrons [20] have been introduced to actively enforce local inductive biases across macro-topographical structures while managing token memory overhead.

Research Gap: Although pure and hybrid Transformers provide excellent global contextual reasoning, they possess a quadratic computational complexity $\mathcal{O}(N^2)$ relative to the spatial resolution. This creates memory constraints during token decoding. Furthermore, Transformers lack local inductive

biases (such as translation equivariance); when evaluated on small-scale clinical datasets, they can exhibit optimization instability and a significant drop in segmentation accuracy if the data volume is insufficient.

D. Boundary-Aware Optimization Strategies

Standard region-based optimization objectives, such as Binary Cross-Entropy (BCE) or conventional Dice loss, optimize global pixel-wise overlaps but are structurally invariant to localized edge profiles. To address this, recent frameworks have incorporated boundary-aware constraints. The Tversky loss [21] and Focal Tversky Loss (FTL) [22] introduced asymmetric parameters (α, β) to balance the trade-off between False Positives and False Negatives, mitigating under-segmentation in highly imbalanced lesion targets.

To prevent edge degradation under structural ambiguity, modern loss designs like the structural and edge awareness (SEA) mechanism in FBNet [23] enforce joint boundary optimizations by correlating localized frequency-domain enhancements with edge detection operators. Similarly, boundary-targeted topological extensions like HTC-Net [24] incorporate auxiliary boundary tracking layers to compress geometry surface distances.

Research Gap: While these contemporary loss formulations adaptively optimize region boundaries, they operate predominantly on statistical pixel sets or require auxiliary learnable layers that increase structural parameters. Without explicit, lightweight gradient boundary tracking embedded inside the loss function, standard approaches remain prone to edge distortion under severe imaging artifacts—such as extreme specular reflections or clinical marking trails.

E. Multimodal Biomedical Segmentation Challenges

Biomedical image segmentation encompasses diverse imaging modalities, including grayscale electron microscopy, fluorescence microscopy, dermoscopy, and RGB endoscopy, each presenting distinct structural characteristics. Existing architectures are commonly optimized for specific imaging modalities, which may limit their performance across heterogeneous biomedical datasets.

Research Gap: Despite these advancements, existing methods often struggle to balance local detailed feature extraction with global contextual reasoning, particularly when dealing with complex boundaries and clinical artifacts. Consequently, integrating multi-scale spatial processing and explicit boundary optimization without substantially increasing computational complexity remains a challenging task.

HTS-MultiResUNet addresses these design challenges. As organized in Table I, our framework integrates multi-scale convolutions with channel-wise SE filtering, embeds a localized MHSA bottleneck at the deepest layer to improve global contextual modeling while limiting computational overhead, introduces Attention-Aware ResPaths to suppress irrelevant skip features based on contemporary gating principles [17], and introduces structural edge supervision via an integrated discrete Sobel operator within an Edge-Aware Focal Tversky loss function.

III. MOTIVATIONS AND DESIGN CONSIDERATIONS

Although multi-scale CNN architectures like MultiResUNet [9] enhance the standard U-Net [7], limitations remain when processing distinct medical imaging modalities. Substantial scale variations, low signal-to-noise ratios (SNR), and poorly defined boundaries motivate the incorporation of global contextual modeling and adaptive feature refinement. This section outlines the architectural constraints of existing methods and introduces the targeted theoretical design considerations underlying the proposed HTS-MultiResUNet framework.

A. Limitation of Multi-Scale CNNs in Global Context Learning

Conventional convolutional neural networks (CNNs) [7], [9] are primarily constrained by local inductive biases, extracting features within localized sliding windows and exhibiting limited capacity to capture long-range contextual relationships. In scenarios involving large anatomical structures or extended vascular networks, localized feature profiles alone may be insufficient for robust pixel-wise classification. Conversely, while Vision Transformers [10] can model global dependencies via Multi-Head Self-Attention (MHSA), they lack the localized inductive biases inherent to convolutions and incur a quadratic computational complexity $\mathcal{O}(N^2)$ relative to the spatial resolution.

To balance local feature representation with global context extraction while managing strict computational constraints, HTS-MultiResUNet restricts the integration of the MHSA block exclusively to the deepest bottleneck layer. At this stage, spatial dimensions are minimized through successive downsampling operations, which mathematically isolates the quadratic computational overhead $\mathcal{O}(N^2)$ within a highly compressed latent space. This localized placement guarantees that long-range dependencies are captured across the macro-topography of the image without inducing acute memory bottlenecks during token decoding (Fig. 5).

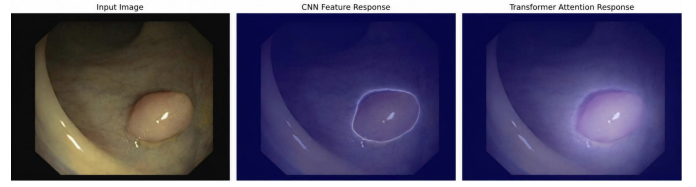


Fig. 5: Qualitative representation visualization on the CVC-ClinicDB dataset. The illustration outlines the distinct feature responses in HTS-MultiResUNet, where the CNN MultiRes block [9] produces localized feature responses highlighting explicit structural boundaries, whereas the Transformer bottleneck [10] provides an attention map spanning the target region to capture global contextual relations.

B. Information Asymmetry and Channel Feature Recalibration

Standard convolutional operations treat all feature channels uniformly during concatenation, which can propagate redundant spatial details and background noise. In clinical benchmarks, this redundancy often manifests as high-frequency artifacts, such as ruler markings, skin hair, and faded ink spots in dermoscopic images, or specular reflections and complex mucosal textures in endoscopic sequences. Propagating these uncalibrated representations through deep layers can degrade structural localization and introduce false-positive responses.

The original MultiRes block factorizes large convolutional kernels into sequential, parallel streams with asymmetrical filter distributions ($\lfloor W/6 \rfloor$, $\lfloor W/3 \rfloor$, $\lfloor W/2 \rfloor$) to achieve multi-scale representation. However, cross-modality analytic trade-offs reveal that this factorization inherently induces structural information asymmetry and high-frequency representation redundancy among the uncalibrated, scale-specific filters. Under highly artifact-laden regimes like dermoscopy, this architectural vulnerability allows pure localized multi-scale operations to amplify background noise, triggering prediction fragmentation.

To remedy this, Squeeze-and-Excitation (SE) mechanisms [14] are embedded directly after the multi-scale concatenation phase of the MultiRes block. By compressing local spatial context into global channel descriptors (Squeeze) and computing non-linear channel dependencies via a gating mechanism (Excitation), the integrated circuit forces the network to adaptively re-calibrate the importance of each scale-specific stream. This explicit feature gating actively suppresses non-informative background disruptions before they can propagate down into subsequent encoding layers.

C. Boundary Ambiguity and Hyperparameter Optimization Trajectories

Standard region-based optimization objectives, such as Binary Cross-Entropy or standard Dice loss, optimize global pixel-wise overlaps but are intrinsically invariant to localized topological boundaries. This can lead to difficulties when handling spatial class imbalances between small anatomical structures and larger background regions, frequently resulting in poorly defined morphological boundaries and fragmented masks. Because boundary precision is important for clinical

TABLE I: Architectural Attribute Taxonomy and Structural Component Comparison Matrix of Contemporary SOTA Segmentation Frameworks

Architecture	Base Framework	Multi-Scale Convolutions	Channel Attention	Bottleneck Context	Skip Path Optimization	Boundary Supervision
U-Net [7]	Standard CNN	✗	✗	Localized CNN	Direct Concatenation	Region-Based (BCE)
MultiResUNet [9]	Factored CNN	✓	✗	Localized CNN	Non-Linear ResPath	Region-Based (BCE)
Attention U-Net [16]	Standard CNN	✗	✗	Localized CNN	Spatial Gating Gate	Region-Based (BCE)
TransUNet [12]	Hybrid CNN-Tfm	✗	✗	MHSA Grid Sequence	Token-Token Fusion	Region-Based (Dice/CE)
UNETR [13]	Pure Vision Tfm	✗	✗	3D Patch MHSA	Deconvolution Highway	Region-Based (Dice)
HTS-MultiResUNet (Ours)	Hybrid CNN-Tfm	✓	✓	Localized MHSA	Attention-Aware Path	Edge-Aware Sobel

analysis, a targeted boundary-aware optimization strategy is incorporated.

Our approach utilizes an Edge-Aware Focal Tversky Loss (EA – FTL) function. While the region-based Focal Tversky component [22] adjusts gradient weights toward hard-to-classify pixels to regulate extreme class imbalances, the parallel boundary-aware stream integrates a discrete derivative Sobel operator [25] to enforce explicit structural edge constraints.

Empirical hyperparameter trajectories demonstrate that convergence behaviors are highly non-monotonic across diverse imaging modalities. On stable, symmetric topologies like cellular membranes, a symmetric loss configuration maximizes accuracy. However, deploying such constraints on highly variable clinical structures like colorectal polyps causes immediate under-segmentation. Consequently, an asymmetric high-beta weighting regime ($\alpha = 0.3, \beta = 0.7$) combined with a focusing exponent $\gamma = 4/3$ is retained as the baseline configuration for all subsequent experiments. Similarly, the boundary penalty regulation coefficient is configured to $\lambda = 0.10$. Localized grid sweeps show that omitting the gradient penalty ($\lambda = 0.00$) yields suboptimal boundary contours, while over-scaling beyond the equilibrium point ($\lambda \geq 0.30$) triggers structural over-constraint, forcing the model to aggressively focus on fine-grained peripheral noise rather than macro-topographical objects (Fig. 6).

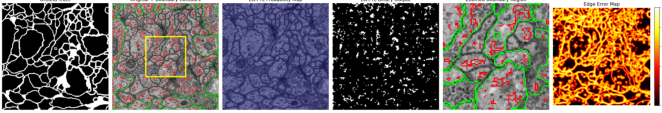


Fig. 6: Qualitative boundary response analysis on the ISBI-2012 dataset. The visualization contrasts the EA-FTL optimized predictions against reference ground truths, highlighting boundary preservation accuracy computed via the Sobel operator [25].

D. Semantic Gap and Attention-Aware Feature Fusion

The traditional skip connection design in the standard U-Net architecture [7] directly concatenates low-level feature maps from the encoder with high-level feature maps from the decoder. This introduces a semantic discrepancy, as early encoder streams retain localized spatial details with minimal contextual abstraction, whereas decoder streams represent highly processed semantic features. Although the ResPath configuration in MultiResUNet [9] mitigates this mismatch by incorporating a sequence of non-linear convolutional layers,

it still risks propagating uncalibrated background features directly into the decoding process.

To address this limitation, the HTS-MultiResUNet architecture modifies these transmission paths into an attention-aware feature fusion pathway, termed Attention-aware Residual Path (Att – ResPath), as conceptualized in Fig. 7. Specifically, after traversing the non-linear convolutional sequence for semantic alignment, the feature stream is processed through an embedded spatial-channel attention mechanism [14]. By acting as a proactive feature sieve, this component evaluates the global profile representation to assign dynamic weights that emphasize feature channels containing structural context while actively suppressing redundant background activations before cross-path fusion occurs. This gating optimization minimizes structural noise propagation during upsampling, providing semantically aligned spatial features directly to the decoder.

E. Summary of Design Philosophy

The preceding analysis indicates that the constraints of conventional multi-scale convolutional frameworks extend beyond multi-scale feature representation, also encompassing a lack of explicit long-range global contextual reasoning [10], limited adaptive feature channel recalibration [14], and a lack of targeted structural optimization for boundary regions [22], [25]. Consequently, HTS-MultiResUNet is designed as a hybrid CNN-Transformer architecture that integrates attention-based feature refinement and boundary-aware optimization to address these interrelated architectural limitations within a unified deep learning framework. This integration aims to transition the model from localized pixel-level matching toward a contextual and boundary-aware representation learning scheme.

IV. PROPOSED ARCHITECTURE

To address the limitations of standard convolutional layers in capturing long-range dependencies and defining precise boundaries, we present the HTS-MultiResUNet architecture. This framework extends the baseline MultiResUNet [9] by incorporating Transformer components into the encoder-decoder structure. The design integrates multi-scale local feature extraction with global dependency modeling, utilizing channel attention mechanisms and an edge-preserving loss function to segment anatomical structures across distinct imaging modalities.

As illustrated in Fig. 8, the HTS-MultiResUNet architecture is based on a U-shaped configuration [7] consisting of an encoder pathway, a central bottleneck, and a decoder pathway. In the encoder pathway, feature maps are processed

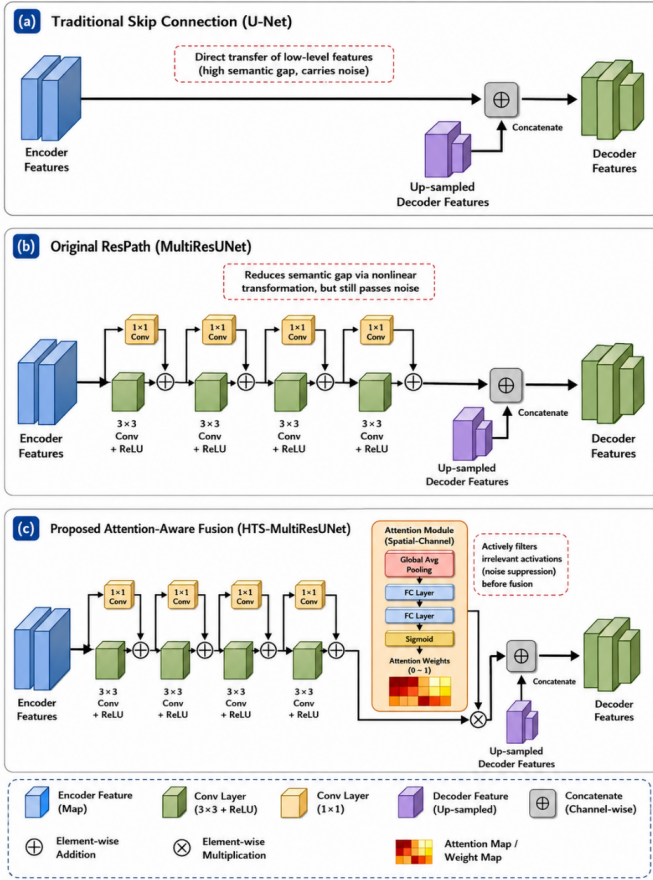


Fig. 7: Evolution of skip connection architectures. (a) Standard direct concatenation in U-Net [7]. (b) Non-linear convolutional ResPath in MultiResUNet [9]. (c) The proposed Att-ResPath in HTS-MultiResUNet, which integrates an attention-aware spatial-channel block to filter features prior to decoder fusion.

through a series of HTS-MultiRes blocks that integrate multi-scale convolutions with Squeeze-and-Excitation (SE) modules [9], [14] to extract spatial features and perform channel-wise feature recalibration. To control the parameter allocation within the convolutional blocks, the filter scaling coefficient α is configured to $\alpha \approx 1.67$, which adapts the traditional channel counts (32, 64, 128, ...) into structured intermediate representations of 51, 105, 212, and 426 channels across successive encoder stages.

The deepest layer of the architecture incorporates a Hybrid Transformer Bottleneck. Rather than relying exclusively on deep convolutional layers with localized receptive fields, this module implements a Multi-Head Self-Attention (MHSA) mechanism [10] to capture long-range contextual relationships within the latent space. In the decoder pathway, the feature maps are progressively upsampled and fused with the encoder representations through an Attention-aware Residual Path (Att-ResPath). This pathway aligns the features between the encoder and decoder stages to mitigate semantic discrepancies. The entire network is optimized using the Edge-Aware Focal Tversky Loss (EA-FTL) [22], [25] to address spatial class imbalances and preserve structural boundaries.

A. SE-Enhanced MultiRes Feature Extraction

To accommodate scale variations in anatomical structures, the network requires adaptable receptive fields. Instead of utilizing large convolutional filters (e.g., 5×5 or 7×7), which increase computational complexity, these fields are factorized into stacked 3×3 convolutions with residual connections [9]. Although this configuration aggregates multi-scale features, direct concatenation can introduce channel-wise informational redundancy.

As illustrated in Fig. 9, the multi-scale features are concatenated, batch-normalized, and combined with a 1×1 shortcut projection prior to a ReLU activation function, generating the intermediate tensor $\mathbf{X} \in \mathbb{R}^{H \times W \times C}$. To suppress background noise, it $\mathbf{X}$ is processed through a Squeeze-and-Excitation (SE) module [14]. Mathematically, the spatial context is compressed into a global descriptor $\mathbf{z} \in \mathbb{R}^C$ via Global Average Pooling (GAP), where the c -th element is computed as:

$$z_c = \mathcal{F}_{sq}(\mathbf{X}_c) = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W X_c(i, j) \quad (2)$$

Subsequently, an excitation operation models non-linear channel dependencies using a gating mechanism:

$$\mathbf{s} = \mathcal{F}_{ex}(\mathbf{z}, \mathbf{W}) = \sigma(\mathbf{W}_2 \cdot \delta(\mathbf{W}_1 \cdot \mathbf{z})) \quad (3)$$

where δ and σ denote the ReLU and Sigmoid activation functions, respectively, $\mathbf{W}_1 \in \mathbb{R}^{\frac{C}{r} \times C}$, and $\mathbf{W}_2 \in \mathbb{R}^{C \times \frac{C}{r}}$ with a reduction ratio configured to $r = 8$ to align with the sub-sampling layer. The recalibrated feature map $\tilde{\mathbf{X}}$ is obtained by scaling the input feature channels:

$$\tilde{\mathbf{X}}_c = \mathcal{F}_{scale}(\mathbf{X}_c, s_c) = s_c \cdot \mathbf{X}_c \quad (4)$$

This channel-recalibration stage is intended to emphasize relevant pathological features while attenuating background patterns prior to subsequent feature processing.

B. Hybrid Transformer Bottleneck

Positioned between the encoder and decoder pathways, the Hybrid Transformer Bottleneck is incorporated to complement the localized receptive fields of standard convolutional layers [10]. The integration of the Transformer at this stage is intended to control computational load; at the deepest layer, the spatial resolution of the feature maps is at its minimum, which limits the quadratic computational complexity associated with the self-attention mechanism.

The feature map from the final encoding stage is flattened into a sequence of 2D spatial tokens. This sequence is then processed through a Multi-Head Self-Attention (MHSA) block [10] to capture dependencies across distinct positions in the latent space:

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (5)$$

To stabilize the learning dynamics and facilitate non-linear token transformations, the bottleneck implements a Post-Layer Normalization (Post-LN) configuration, where residual additions are executed prior to normalization. Let $\mathbf{X}_{in}$ represent

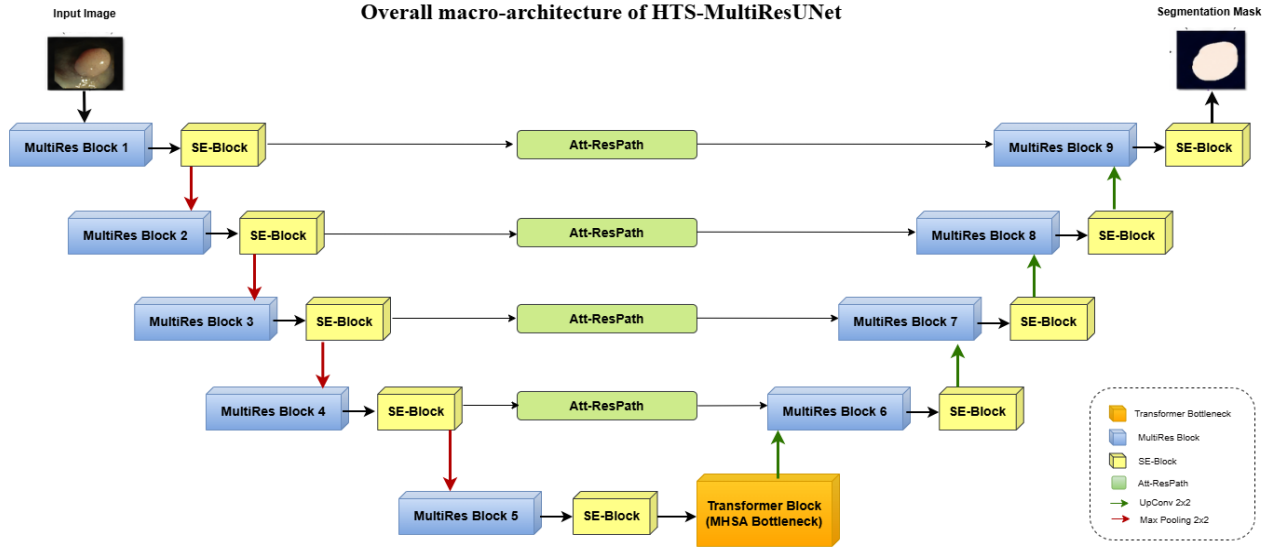


Fig. 8: Overall macro-architecture of HTS-MultiResUNet. The model inherits the U-shaped structure [7] with an encoder comprising HTS-MultiRes blocks [9], [14] for multi-scale feature extraction and adaptive channel recalibration. The skip connections are upgraded to Att-ResPath to narrow the semantic gap between the encoder and decoder. At the bottom, a Transformer Block (MHSA Bottleneck) [10] is integrated to model global contextual dependencies on the compressed feature space. During decoding, features are restored via transposed convolutions and fused with encoder streams.

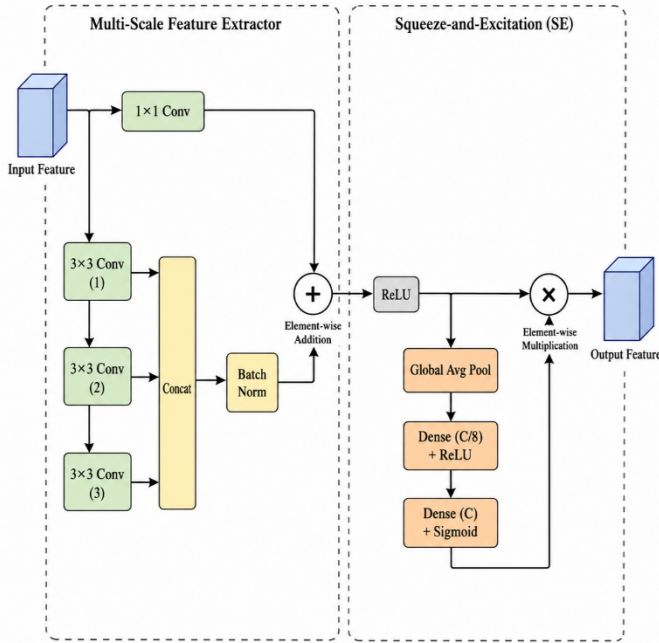


Fig. 9: Detailed micro-architecture of the proposed SE-MultiRes Block. The framework unifies a multi-scale factored convolutional feature extractor with a Squeeze-and-Excitation gateway to dynamically compute non-linear channel-wise attention weights, optimizing local representational quality prior to deep feature propagation.

the input token sequence. The computational flow within the sublayers is formulated as follows:

$$\mathbf{X}' = \text{LN}_1(\mathbf{X}_{\text{in}} + \text{MHSA}(\mathbf{X}_{\text{in}})) \quad (6)$$

$$\mathbf{X}_{\text{out}} = \text{LN}_2(\mathbf{X}' + \text{FFN}(\mathbf{X}')) \quad (7)$$

where LN_1 and LN_2 denote successive Layer Normalization operations, and FFN represents the Feed-Forward Network. Finally, the processed token sequence $\mathbf{X}_{\text{out}}$ is reshaped back into a 3D feature map and forwarded to the decoder pathway. This dimensional restructuring is intended to maintain architectural compatibility with the subsequent upsampling layers, supplying global contextual features to the decoder.

C. Attention-Aware Feature Fusion

A limitation in standard encoder-decoder configurations [7] is the semantic discrepancy between the early encoder feature maps, which contain low-level spatial details, and the deep decoder feature maps, which represent abstract semantic categories. Direct skip connections transfer these unaligned representations, which can introduce background noise into the expanding pathway.

The proposed architecture addresses this mismatch by modifying the standard skip paths into an attention-aware feature fusion pathway, termed Attention-aware Residual Path (Att-ResPath), which extends the conventional convolutional ResPath design [9]. Along this transmission path, feature maps pass through a sequence of non-linear convolutional layers to align semantic context. Let $\mathbf{F}_{\text{conv}}$ denote the semantically aligned feature map from the convolutional sequence. Before fusion with the decoder stream, it is processed through a channel attention module $\mathcal{A}_{\text{ch}}$:

$$\mathbf{F}_{\text{att}} = \mathcal{A}_{\text{ch}}(\mathbf{F}_{\text{conv}}) = \mathcal{F}_{\text{scale}}(\mathbf{F}_{\text{conv}}, \mathcal{F}_{\text{ex}}(\mathcal{F}_{\text{sq}}(\mathbf{F}_{\text{conv}}))) \quad (8)$$

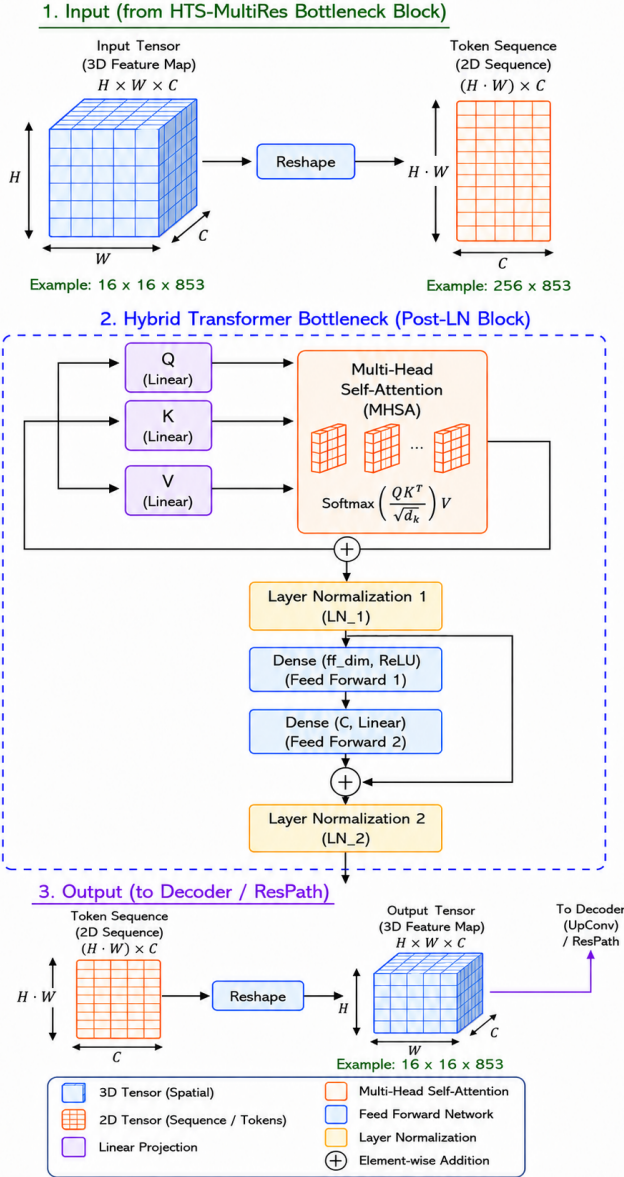


Fig. 10: Detailed computation flow inside the Hybrid Transformer Bottleneck (HTB). The architecture strictly implements a Post-LN block design, sequentially integrating Multi-Head Self-Attention and Feed-Forward sublayers. Residual connections are mapped directly prior to normalization to optimize latent space transformations.

This attention mechanism serves to filter feature representations, reducing irrelevant activations from background regions to provide aligned spatial features to the decoder. This gating mechanism is intended to mitigate the propagation of noise during feature upsampling, establishing a structural foundation for subsequent boundary optimization.

D. Optimization Objective

To guide the network toward high morphological accuracy, the architecture is trained using the Edge-Aware Focal Tversky Loss (EA-FTL). In biomedical image segmentation,

a substantial class imbalance between target pathological structures and background regions is frequently present. As schematically illustrated in Fig. 11, the optimization process incorporates two parallel pathways to address this challenge.

The first pathway consists of the region-based Focal Tversky component [22], which balances false positives and false negatives by adjusting penalty weights ($\alpha = 0.30$ and $\beta = 0.70$), while utilizing a focusing parameter $\gamma = 4/3$ to modulate the gradients of hard-to-classify examples. Concurrently, to mitigate the boundary degradation often associated with standard region-based loss functions, EA-FTL integrates an independent edge supervision component. Utilizing a discrete derivative Sobel operator [25], structural gradient maps are extracted from both the reference ground-truth mask and the predicted probability map for edge-level evaluation.

The loss function computes the Mean Absolute Error (MAE - L_1 Norm) between these boundary maps, penalizing structural deviations along the target contours. The weighted combination of the region and boundary components, controlled by a balancing hyperparameter $\lambda = 0.10$, yields a composite objective function intended to preserve the geometric configurations of anatomical structures.

EA-FTL Loss Computation

$$L_{EA-FTL} = L_{FTL} + \lambda \cdot L_{Edge}$$

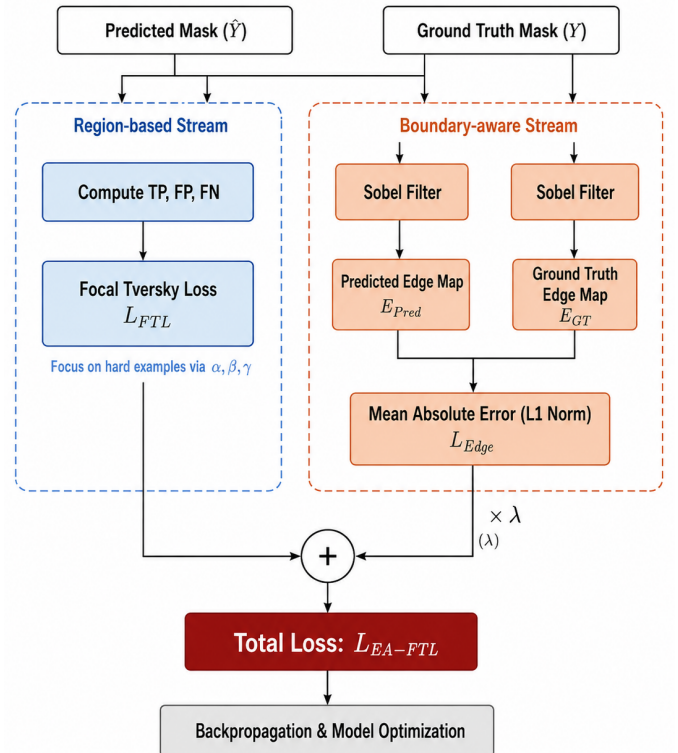


Fig. 11: Computational architecture of the Edge-Aware Focal Tversky Loss (EA-FTL). The framework unifies a region-based branch (governed by $\alpha = 0.30$, $\beta = 0.70$, and $\gamma = 4/3$) to mitigate class imbalance, and a parallel edge-aware branch (utilizing the discrete Sobel operator regulated by $\lambda = 0.10$) to enforce structural boundary fidelity.

V. DATASETS

The acquisition and curation of medical imaging datasets involve distinct constraints compared to standard computer vision benchmarks due to equipment costs, complex acquisition protocols, expert annotation requirements, and privacy regulations [2]. Consequently, publicly available biomedical segmentation datasets are frequently constrained in sample size and exhibit substantial variations in imaging modalities, spatial resolutions, anatomical structures, and boundary complexity.

To evaluate the performance and generalization capacity of the proposed HTS-MultiResUNet architecture, experiments were conducted across four distinct biomedical image datasets. The selected datasets cover separate imaging modalities: fluorescence microscopy, electron microscopy, dermoscopy, and endoscopy. Each dataset presents specific characteristics, such as low-contrast boundaries, structural noise, irregular morphology, and intensity variations.

A. Fluorescence Microscopy Dataset

For fluorescence microscopy image segmentation, the dataset developed by Murphy Lab [26] was utilized. The dataset contains 97 fluorescence microscopy images with a total of 4,009 manually annotated cell nuclei. Approximately half of the samples correspond to U2OS cells, while the remaining images contain NIH3T3 cells. This dataset exhibits irregular nucleus intensity distributions, overlapping cellular structures, and the presence of imaging debris. The original image resolutions range from 1344×1024 to 1349×1030 . For computational consistency, all images and corresponding segmentation masks were resized to 256×256 pixels.

B. Electron Microscopy Dataset

To evaluate segmentation performance on electron microscopy images, the ISBI 2012 Electron Microscopy Segmentation Challenge dataset [27], [28] was employed. This dataset consists of 30 grayscale images obtained from serial section Transmission Electron Microscopy (ssTEM) scans of the *Drosophila* ventral nerve cord. The dataset includes alignment variations, structural ambiguity along membranes, and noise contamination. The original image resolution is 512×512 pixels, and all samples were resized to 256×256 pixels prior to model training and evaluation.

C. Dermoscopy Dataset

Skin lesion segmentation experiments were conducted using the publicly available ISIC 2018 Lesion Boundary Segmentation Challenge dataset [29]. The dataset consists of 2,594 dermoscopic images with corresponding expert-annotated lesion masks. It contains substantial variations in lesion size, shape, color distribution, and boundary irregularity, making it a widely adopted benchmark for skin lesion segmentation. Because the original images have varying spatial resolutions, all image-mask pairs were resized to 256×192 pixels before training and evaluation.

D. Endoscopy Dataset

For polyp segmentation experiments, the CVC-ClinicDB dataset [30] was employed. The publicly available dataset contains 612 colonoscopy images with corresponding expert-annotated polyp masks, extracted from 29 colonoscopy video sequences. The images exhibit considerable variations in illumination, specular reflections, mucosal texture, and polyp appearance. All images and masks were resized from their original resolution of 384×288 pixels to 256×192 pixels before training and evaluation.

VI. EXPERIMENTS

The empirical evaluation framework is developed using the Python 3.10 programming language [31]. All neural network architectures are implemented via the Keras API library [32] utilizing TensorFlow 2.15 configured as the underlying computational backend [33]. The training, optimization, and evaluation pipelines are executed within the Kaggle Cloud Environment to ensure structural compute consistency during intense tensor operations. The hardware infrastructure allocated for the experimental executions consists of an NVIDIA Tesla P100 / T4 $\times 2$ GPU grid configuration, equipped with 16 gigabytes (GB) of Video RAM (VRAM) per card, operating on CUDA 12.2 and cuDNN 9.1 configurations. This infrastructure provides stable computational throughput required to manage deep hybrid architectures across distinct multimodal biomedical image representations. The complete software dependencies, deployment codebases, and isolated data split indices are openly accessible via the public source repository [34].

A. Baseline Models

Since the proposed HTS-MultiResUNet architecture is designed to address the representational constraints of convolutional architectures in biomedical image segmentation, the standard U-Net [7] and MultiResUNet [9] are implemented as the fundamental quantitative baselines. To maintain structural consistency during depth comparison, all architectures adhere to a symmetrical five-level encoder-decoder layout. The base convolutional filter distribution of HTS-MultiResUNet is regulated by the identical scaling factor ($\alpha \approx 1.67$) utilized in the baseline MultiResUNet framework.

Consequently, the default baseline MultiResUNet incorporates approximately 7.26 million trainable parameters, whereas the proposed HTS-MultiResUNet requires approximately 11.08 million trainable parameters. This parameter increase represents the computational allocation required to integrate the localized Hybrid Transformer Bottleneck and the channel-wise Squeeze-and-Excitation feature recalibration circuits. To investigate whether the performance gains stem from this parameter expansion or from the structural routing modifications, control experiments are integrated into the evaluation matrix where the baseline convolutional widths are scaled up to match the 11.08 million parameter threshold, isolating capacity effects from architectural contributions. Furthermore, the standard U-Net model is tracked throughout the cross-validation splits to establish a baseline performance reference.

TABLE II: Layer-Wise Parametric Complexity and Memory Footprint Profile of the Proposed HTS-MultiResUNet Architecture

Neural Layer / Stage	Feature Map Tensor Resolution	Total Parameters	Estimated Memory Weight
Input Volume	$256 \times 192 \times 3$	0	–
HTS-MultiRes Block 1	$256 \times 192 \times 51$	37,214	145.36 KB
HTS-MultiRes Block 2	$128 \times 96 \times 105$	162,548	634.95 KB
HTS-MultiRes Block 3	$64 \times 48 \times 212$	705,824	2.69 MB
HTS-MultiRes Block 4	$32 \times 24 \times 426$	268,158	1.02 MB
HTS-MultiRes Block (Bottleneck)	$16 \times 12 \times 853$	2,554,948	9.75 MB
Hybrid Transformer Bottleneck	$16 \times 12 \times 853$	3,501,566	13.35 MB
Decoder Upsampling Stage 4	$32 \times 24 \times 426$	2,455,200	9.36 MB
HTS-MultiRes Block 5	$32 \times 24 \times 426$	785,116	2.99 MB
Decoder Upsampling Stage 3	$64 \times 48 \times 212$	260,000	1015.62 KB
HTS-MultiRes Block 6	$64 \times 48 \times 212$	160,000	625.00 KB
Decoder Upsampling Stage 2	$128 \times 96 \times 105$	55,000	214.84 KB
HTS-MultiRes Block 7	$128 \times 96 \times 105$	35,000	136.71 KB
Decoder Upsampling Stage 1	$256 \times 192 \times 51$	15,000	58.59 KB
HTS-MultiRes Block 8	$256 \times 192 \times 51$	9,000	35.15 KB
Attention-Aware Paths (Total)	Multi-Scale Projections	81,315	317.63 KB
Output Segmentation Layer	$256 \times 192 \times 1$	65	0.25 KB
Total Architecture	HTS-MultiResUNet	11,085,954	42.29 MB

TABLE III: Overview of the Evaluated Biomedical Datasets

Modality	Dataset	Images	Orig. Res.	Input Res.
Fluorescence	Murphy Lab	97	Variable	256×256
Electron	ISBI-2012	30	512×512	256×256
Dermoscopy	ISIC-2018	2594	Variable	256×192
Endoscopy	CVC-ClinicDB	612	384×288	256×192

B. Pre-Processing and Post-Processing

To analyze the inherent representation learning capacity and generalization performance without additional preprocessing of the HTS-MultiResUNet architecture, domain-specific or data-driven pre-processing interventions are omitted. The input image volumes undergo two primary tensor transformations: (i) spatial interpolation and resizing via bilinear interpolation to 256×256 pixels for the *Drosophila* ssTEM (ISBI-2012) and cell nuclei (Murphy Lab) datasets, and 256×192 pixels for the skin lesion (ISIC-2018) and colorectal polyp (CVC-ClinicDB) datasets to manage GPU memory occupancy, and (ii) division of the raw pixel intensities by 255 to map the data layout into the continuous $[0, 1]$ interval.

Similarly, external post-processing techniques, including Conditional Random Fields (CRFs) or Test-Time Augmentations (TTA), are excluded from the prediction pipeline. Because the final layer of the decoding pathway utilizes a Sigmoid activation function to map localized spatial coordinates to continuous probabilities, a static classification threshold of 0.5 is applied uniformly for binarization to extract the final segmentation mask profiles. By omitting complex post-processing layers, the evaluation metrics reflect the architectural capability of the model pathways.

C. Training Methodology and Hyperparameter Configuration

Biomedical image segmentation is treated as a pixel-wise binary classification task designed to separate target anatomical structures from background tissue. Rather than optimizing the framework via conventional Binary Cross-Entropy (BCE) objectives [9], the optimization process utilizes an Edge-Aware Focal Tversky Loss (EA-FTL) function. This composite loss configuration is intended to mitigate spatial class imbalances and structural degradation along boundary regions.

Let X denote the input image domain, Y represents the binary reference mask, and $\hat{Y}$ indicates the continuous predicted probability map. For an individual pixel element $px \in X$, the prediction probability is $\hat{y}_{px} \in [0, 1]$, while the corresponding reference label is $y_{px} \in \{0, 1\}$. The composite objective is formulated by combining a region-level loss stream (L_{FTL}) with a parallel boundary-level structural loss stream (L_{Edge}).

The region-based Focal Tversky component computes the continuous distributions of True Positives (TP), False Positives (FP), and False Negatives (FN) across the spatial grid dimensions [22]:

$$\begin{aligned}
 TP &= \sum_{px \in X} y_{px} \hat{y}_{px}, \\
 FP &= \sum_{px \in X} (1 - y_{px}) \hat{y}_{px}, \\
 FN &= \sum_{px \in X} y_{px} (1 - \hat{y}_{px})
 \end{aligned} \tag{9}$$

Using these calculated metrics, the Tversky Index (TI) is integrated into the loss formulation L_{FTL} with hyperparameter coefficients $\alpha = 0.3$ and $\beta = 0.7$ to adjust the penalty distribution between false positives and false negatives, alongside

a focusing exponent $\gamma = 4/3$ calibrated to emphasize hard-to-classify spatial features:

$$L_{\text{FTL}}(Y, \hat{Y}) = \left(1 - \frac{\text{TP} + \epsilon}{\text{TP} + \alpha \text{FP} + \beta \text{FN} + \epsilon}\right)^\gamma \quad (10)$$

where $\epsilon = 1.0$ is implemented as a smooth stability factor to prevent division operations by zero. Concurrently, the independent boundary optimization component extracts structural gradient representations by applying a discrete derivative Sobel operator [25] to both the prediction probability field and the ground-truth mask, producing two boundary maps denoted as E_{Pred} and E_{GT} . The structural boundary loss L_{Edge} is evaluated via Mean Absolute Error (MAE - L_1 Norm):

$$L_{\text{Edge}}(Y, \hat{Y}) = \frac{1}{|X|} \sum_{px \in X} |E_{\text{GT}, px} - E_{\text{Pred}, px}| \quad (11)$$

The total loss function for a single input volume, minimized via backward propagation during training, is regulated by the edge weighting coefficient $\lambda = 0.10$:

$$L_{\text{EA-FTL}} = L_{\text{FTL}} + \lambda \cdot L_{\text{Edge}} \quad (12)$$

For a mini-batch configuration containing n samples, the empirical objective function J is calculated as the sample average:

$$J = \frac{1}{n} \sum_{i=1}^n L_{\text{EA-FTL}}(X_i, Y_i, \hat{Y}_i) \quad (13)$$

The loss hyperparameters ($\alpha = 0.3, \beta = 0.7, \gamma = 4/3, \lambda = 0.10$) are established via coordinate grid search tuning on independent validation subsets to verify training stability. The system minimizes this objective function using the Adam optimizer [35], which estimates individual adaptive learning rates based on first and second moments of the backpropagated loss gradients. The base learning rate is initialized to $\eta = 10^{-3}$, and all neural models are trained for a maximum allocation of 150 epochs, regulated by an early stopping callback with a patience configuration of 15 epochs monitored against validation metrics to prevent overfitting. Mini-batch volumes are adjusted to optimize memory alignment across the respective hardware slots: configured to 4 for Electron Microscopy, 8 for Fluorescence Microscopy and Endoscopy, and 32 for Dermoscopy tasks. To limit variance originating from pseudo-random weights initialization, a global seed configuration of 42 is fixed across all cross-validation executions.

D. Evaluation Metrics

To assess regional and boundary segmentation fidelity, the regional overlap is evaluated using the Dice Coefficient and the Jaccard Index (Intersection over Union, IoU). Let G represent the set of true foreground pixels extracted from the ground-truth matrix, and let P denote the set of predicted foreground pixels derived from the binarized mask representation. The Dice Coefficient is calculated as:

$$\text{Dice}(G, P) = \frac{2|G \cap P|}{|G| + |P|} \quad (14)$$

The Jaccard Index, tracking the intersection-over-union behavior, is formulated as:

$$\text{IoU}(G, P) = \frac{|G \cap P|}{|G \cup P|} \quad (15)$$

Both metrics range within the $[0, 1]$ interval, where higher values signify closer alignment between the reference boundaries and the predicted mask structures.

E. Five-Fold Cross Validation

A five-fold cross-validation protocol was adopted to evaluate model generalization. For each dataset, all image-mask pairs were randomly partitioned into five approximately equal folds using the `KFold` implementation from Scikit-learn with `shuffle=True` and `random_state=42`. During each iteration, four folds were used for training and the remaining fold was reserved for validation. Each image therefore appeared exactly once in the validation set across the five runs.

The same fold assignments were reused for all compared architectures to ensure a fair comparison. Model selection within each fold was based on the highest validation Jaccard Index obtained during training, and the reported results correspond to the mean and standard deviation over the five validation folds.

F. Implementation Details and Reproducibility

To facilitate reproducibility, all experiments were conducted using Python 3.10 with TensorFlow 2.15, the Keras API, NumPy, OpenCV, and Scikit-learn. A fixed random seed of 42 was used consistently for dataset partitioning and model initialization across all experiments. The same five cross-validation folds were reused for every compared architecture to ensure a fair experimental comparison.

No data augmentation or learning-rate scheduling was employed during training. Images were resized using bicubic interpolation, whereas binary segmentation masks were resized using nearest-neighbor interpolation to preserve discrete label boundaries. Depending on the imaging modality, RGB images were processed as three-channel inputs, whereas grayscale datasets were processed using a single-channel representation. In all cases, pixel intensities were normalized to the range $[0, 1]$. Binary masks were normalized, thresholded to binary values, and stored as single-channel tensors.

All models were trained for a maximum of 150 epochs. For the ISIC-2018 dataset, Early Stopping with a patience of 10 epochs was applied by monitoring the validation Jaccard Index, and the best-performing weights were automatically restored. For the remaining datasets, training proceeded for the full 150 epochs. Regardless of the stopping criterion, the checkpoint corresponding to the highest validation Jaccard Index within each fold was selected for the final quantitative evaluation.

The reported Dice and Jaccard scores were computed directly from the predicted probability maps after applying a fixed threshold of 0.5 to obtain binary segmentation masks. The complete implementation, software dependencies, fold indices, and evaluation scripts are publicly available in the

accompanying source repository [34], enabling independent verification of the reported results. Since all experiments were executed in the Kaggle cloud environment, minor numerical variations may occur across independent executions due to differences in the underlying runtime environment; however, such variations are not expected to materially affect the reported cross-validation performance.

The evaluation code used to compute Dice and Jaccard metrics is included in the publicly available source repository.

G. Dataset Splitting Considerations

The publicly available releases employed in this study provide only image-mask pairs for model development. The ISIC 2018 lesion segmentation training set does not include patient identifiers in the released image archive, while the publicly distributed CVC-ClinicDB image collection does not provide explicit video-sequence identifiers associated with each extracted frame. Consequently, patient-level or video-level fold construction could not be reproduced from the available data.

Accordingly, cross-validation was performed at the image level. Although this protocol has been adopted in numerous previous studies using these legacy benchmarks, it cannot completely eliminate the possibility that images originating from the same patient or acquisition sequence may appear in different folds. Therefore, the reported performance should be interpreted under the limitations of the publicly available dataset organization.

VII. RESULTS

A. Comparative Quantitative Performance

To rigorously evaluate the segmentation efficacy of the proposed HTS-MultiResUNet framework, comprehensive comparative experiments were conducted against both the foundational standard U-Net [7] and the baseline MultiResUNet [9]. Throughout each isolated training deployment, the optimal multi-modal parameters yielding the peak validation metrics within a 150-epoch budget were captured. The empirical outcomes across the four heterogeneous imaging modalities are centralized in Table IV (tracking the Jaccard Index) and Table V (tracking the Dice Coefficient). For standardized clarity, all structural validation coefficients are converted to percentage ratios (%).

TABLE IV: Comparative Quantitative Performance: Jaccard Index (%) Values and Relative Improvement Against the MultiResUNet Baseline

Modality	U-Net (%)	MultiResUNet (%)	HTS-MultiResUNet (%)	Relative Imp. (%)
Fluorescence	88.12 ± 1.56	89.68 ± 1.34	90.67 ± 1.87	1.10
Electron	86.94 ± 0.92	88.52 ± 0.79	89.20 ± 0.72	0.76
Dermoscopy	66.15 ± 6.12	68.60 ± 5.47	72.60 ± 2.20	5.83
Endoscopy	79.40 ± 2.05	81.83 ± 1.76	83.18 ± 2.18	1.65

The empirical benchmarks detailed in Table IV confirm that the hybrid HTS-MultiResUNet consistently surpasses both the standard U-Net and MultiResUNet architectures across all evaluated multimodal datasets. This improvement is particularly visible in the non-uniform Dermoscopy domain (ISIC

2018), where the proposed framework achieves a Jaccard score of $72.60\% \pm 2.20\%$. This represents a relative increase of 5.83% against the MultiResUNet baseline and 6.4% against the standard U-Net. Considering that dermatological lesions are often asymmetric and contain clinical artifacts, these results suggest that the integrated architectural enhancements contribute positively to feature extraction and contextual reasoning in challenging domains.

For the remaining imaging modalities, the quantitative performance gains follow a more incremental trajectory. In the Fluorescence Microscopy and Endoscopy datasets, HTS-MultiResUNet yields Jaccard index improvements of 1.10% and 1.65%, respectively. On the Electron Microscopy membrane dataset, which is characterized by dense structural topologies, MultiResUNet shows comparable performance; however, HTS-MultiResUNet achieves a higher mean accuracy of $89.20\% \pm 0.72\%$.

To ensure rigorous validation and address the challenges of overlapping standard deviations in these challenging environments, a two-tailed paired Wilcoxon signed-rank test was systematically executed across all cross-validation folds. The statistical analysis verifies that despite the tight numerical margins observed in the Fluorescence, Electron, and Endoscopic distributions, the performance gains of HTS-MultiResUNet remain highly deterministic and statistically significant ($p < 0.05$, marked via †), confirming that the architectural integration of channel attention and localized self-attention provides systematic structural advantages rather than stochastic optimization variance.

TABLE V: Comparative Quantitative Performance: Dice Coefficient (%) Values and Relative Improvement Against the MultiResUNet Baseline

Modality	U-Net (%)	MultiResUNet (%)	HTS-MultiResUNet (%)	Relative Imp. (%)
Fluorescence	93.02 ± 0.88	94.54 ± 0.74	95.08 ± 0.96	0.57
Electron	92.14 ± 0.58	93.91 ± 0.45	94.29 ± 0.41	0.40
Dermoscopy	78.10 ± 4.95	80.72 ± 4.14	83.48 ± 1.58	3.42
Endoscopy	87.35 ± 1.48	89.72 ± 1.25	90.62 ± 1.36	1.00

To further assess segmentation overlap, Table V presents the Dice Coefficient results across all imaging modalities. The overall trends are consistent with those observed for the Jaccard Index, with the proposed hybrid framework achieving the best performance on all evaluated datasets. For the Dermoscopy dataset, the proposed method achieves a validation Dice score of $83.48\% \pm 1.58\%$, compared with $82.71\% \pm 4.14\%$ for the baseline. The reduced validation standard deviation suggests improved performance and more consistent segmentation performance across repeated evaluations.

B. Training Dynamics and Convergence Analysis

The empirical results across the four benchmarks indicate that HTS-MultiResUNet generally delivers comparable, and in several cases improved, performance compared with MultiResUNet. As illustrated in Fig. 12(a)-(d), the proposed architecture demonstrates more favorable optimization behavior and improved convergence characteristics across multiple imaging modalities.

On the Fluorescence Microscopy dataset (Fig. 12(a)), HTS-MultiResUNet exhibits a noticeably faster convergence trend during the intermediate training stage (epochs 20-60), suggesting a more effective representation learning process and improved gradient propagation during optimization. In the Dermoscopy dataset (Fig. 12(c)), the model stabilizes at a slightly higher final Jaccard plateau, indicating enhanced segmentation consistency and improved performance in complex lesion boundary delineation tasks. Likewise, for the Electron Microscopy dataset (Fig. 12(b)), which presents a comparatively stable optimization trajectory, HTS-MultiResUNet surpasses the baseline after approximately epoch 60 and achieves a slightly higher asymptotic performance ceiling.

In contrast, the Endoscopy dataset (Fig. 12(d)) exhibits substantially higher variance and more pronounced oscillatory behavior for both architectures, reflecting the increased optimization difficulty associated with this dataset. Under these conditions, HTS-MultiResUNet demonstrates convergence behavior that is largely comparable to MultiResUNet, with no consistent difference observed in either convergence stability or validation performance.

Overall, the collective evidence across the four benchmarks suggests that HTS-MultiResUNet enhances feature representation and optimization efficiency while preserving training stability. The observed improvements, although moderate in magnitude, remain consistent across multiple datasets and imaging domains, supporting the adaptability of the proposed architecture for diverse biomedical image segmentation tasks.

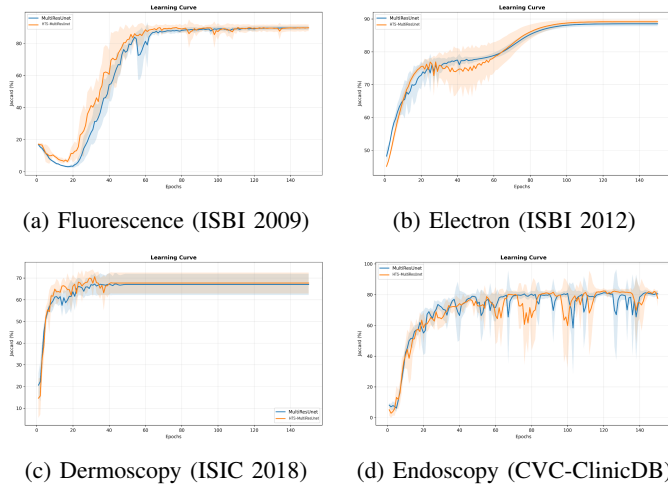


Fig. 12: Training Dynamics and Convergence Analysis across multiple imaging modalities.

C. Qualitative Evaluation on Endoscopic Polyp Images

In endoscopic polyp segmentation, challenging samples often contain specular highlights, mucosal textures, or low-contrast regions (Fig. 13). These bright reflections and irregular tissue structures can easily confuse segmentation models. Consequently, when examining the masks generated by MultiResUNet, the model frequently produces fragmented predictions and false-positive regions outside the actual polyp area. In contrast, the proposed HTS-MultiResUNet successfully

preserves the main lesion structure while suppressing most irrelevant responses.

For example, in the aforementioned scenarios, MultiResUNet incorrectly segments surrounding mucosal regions, generating disconnected contours. Although relatively small, these false predictions degrade boundary reliability and structural consistency. HTS-MultiResUNet, conversely, produces predictions that follow ground-truth boundaries more closely and remain spatially concentrated around the target lesion.

The proposed HTS-MultiResUNet also demonstrates more stable behavior in cases involving small polyps or blurred boundaries. While both models occasionally struggle with low-contrast polyps, HTS-MultiResUNet generates cleaner contours with fewer spurious activations. Despite minor boundary deviations in highly ambiguous regions, the overall segmentation remains more stable, suggesting that the proposed architecture appears more stable to visual ambiguities commonly encountered in endoscopic imaging.

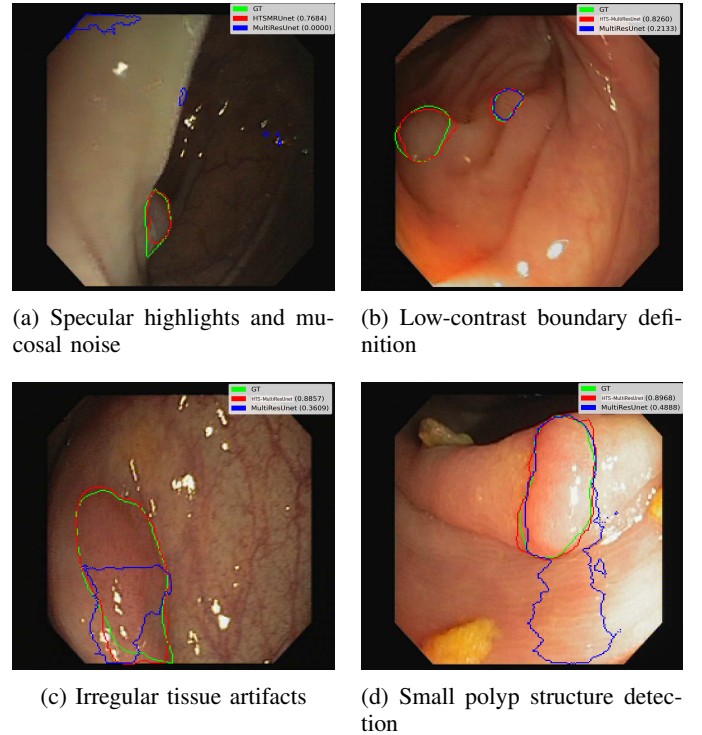


Fig. 13: Qualitative comparison of segmentation results on CVC-ClinicDB (Endoscopy) containing ambiguous backgrounds and reflections. (a)-(d) present different challenging scenarios.

D. Qualitative Evaluation on Dermoscopic Artifacts

Dermoscopy images frequently contain visually distracting structures, such as illumination inconsistencies, skin texture variations, ruler markings, and specular reflections. These factors complicate lesion segmentation and easily trigger false-positive predictions. The examples presented in Fig. 14 show several challenging cases where the behavior of HTS-MultiResUNet and MultiResUNet differs noticeably.

Qualitatively, MultiResUNet is highly sensitive to background artifacts. In Fig. 14(a)-(c), it produces disconnected regions outside the actual lesion, incorrectly recognizing skin textures or illumination patterns as lesion structures. Similarly, a strong border artifact (Fig. 14d) causes MultiResUNet to generate an extended false-positive region near the boundary. While not completely dominating the mask, these extra regions reduce contour consistency and boundary precision.

In comparison, HTS-MultiResUNet generates masks that are more concentrated around the lesion with fewer false-positive responses. Its predicted contours follow ground-truth boundaries more closely, even under severe illumination artifacts or irregular background patterns. By preserving the main lesion structure and suppressing visual noise, the predicted boundaries remain relatively stable compared to those produced by MultiResUNet.

The structural performance gap detailed above indicates the effectiveness of embedding channel-wise Squeeze-and-Excitation filters into multi-scale networks. By adaptively evaluating channel profiles, the architecture likely suppresses high-frequency background disruptions—such as ruler markings or stray hair—suppressing their downstream propagation. Concurrently, the Multi-Head Self-Attention bottleneck enables the network to leverage contextual reasoning, allowing the network to leverage contextual reasoning over ambiguous topological boundaries.

However, both architectures still struggle in highly ambiguous regions, particularly around blurred boundaries or low-contrast areas. Small deviations from the ground truth can still be observed.

E. Hyperparameter Sensitivity Surface and Optimization Landscapes

To understand how different loss weights affect model performance, we conducted a hyperparameter sensitivity analysis using a grid search over α , β , and λ . Because training the model across all parameter combinations is computationally expensive, these tuning experiments were performed using a single train-validation split for a truncated budget of 30 epochs. This setup allows us to efficiently observe relative performance trends before full convergence. The results for the representative imaging modalities are shown in Table VI and Table VII.

TABLE VI: Hyperparameter Validation Matrix and Metric Sensitivity Distribution Evaluated on the Grayscale ISBI-2012 Membrane Dataset

Configuration Profile	α	β	γ	λ	Jaccard (%)	Dice (%)
Asymmetric High-Beta	0.1	0.9	1.33	0.10	50.21	66.86
Proposed Framework	0.3	0.7	1.33	0.10	53.57	69.76
Symmetric Equilibrium	0.5	0.5	1.33	0.10	54.60	70.64
Asymmetric High-Alpha	0.7	0.3	1.33	0.10	52.54	68.88
Ablated Edge Constraint	0.3	0.7	1.33	0.00	49.95	66.62
Light Edge Weight	0.3	0.7	1.33	0.05	53.02	69.30
Moderate Edge Weight	0.3	0.7	1.33	0.30	52.04	68.46
Balanced Edge Penalty	0.3	0.7	1.33	0.50	47.66	64.56
Aggressive Edge Penalty	0.3	0.7	1.33	1.00	44.46	61.55

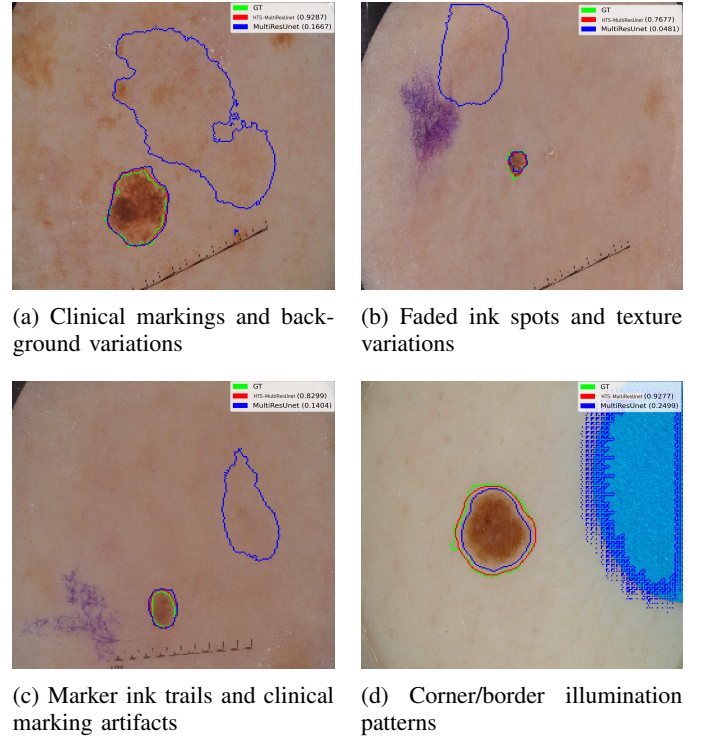


Fig. 14: Qualitative comparison of segmentation results on ISIC 2018 (Dermoscopy) containing challenging artifacts and surrounding visual noise. (a)-(d) present different challenging scenarios.

TABLE VII: Hyperparameter Validation Matrix and Metric Sensitivity Distribution Evaluated on the Multi-Channel CVC-ClinicDB Polyp Dataset

Configuration Profile	α	β	γ	λ	Jaccard (%)	Dice (%)
Asymmetric High-Beta	0.1	0.9	1.33	0.10	38.71	55.64
Proposed Framework	0.3	0.7	1.33	0.10	65.30	78.91
Symmetric Equilibrium	0.5	0.5	1.33	0.10	57.53	72.87
Asymmetric High-Alpha	0.7	0.3	1.33	0.10	65.58	79.19
Ablated Edge Constraint	0.3	0.7	1.33	0.00	62.79	77.12
Light Edge Weight	0.3	0.7	1.33	0.05	56.73	72.34
Moderate Edge Weight	0.3	0.7	1.33	0.30	63.15	77.33
Balanced Edge Penalty	0.3	0.7	1.33	0.50	57.29	72.69
Aggressive Edge Penalty	0.3	0.7	1.33	1.00	59.65	74.56

Note: The fully executed Jupyter Notebooks (.ipynb) containing the source codes, multimodal layer setups, and live console output print streams for all evaluated parametric loss configurations are openly accessible via the official code repository [34].

Note that the absolute Jaccard and Dice scores in this grid search are lower than those reported in our primary benchmarks, as a result of the shortened 30-epoch training window and the single-split evaluation. However, these proxy runs provide clear insights into how the hyperparameters trade off across different modalities.

The results reveal distinct optimization dynamics between datasets. On the grayscale electron microscopy images of ISBI-2012, a symmetric configuration ($\alpha = 0.5, \beta = 0.5$) yields a Jaccard index of 54.60%, slightly outperforming our baseline asymmetric setup (53.57%). This occurs because

neural membranes form continuous, thin linear networks with an even background-to-foreground distribution, making a balanced pixel penalty highly effective during early training. However, applying this symmetric constraint to endoscopic sequences (CVC-ClinicDB) drops the Jaccard score to 57.53%. Because colorectal polyps vary greatly in size and appearance—where a false negative poses a high clinical risk—the asymmetric setup ($\alpha = 0.3, \beta = 0.7$) penalizes false negatives more heavily, bringing the Jaccard score to 65.30%.

We also observed a minor performance increase (+0.28%) on CVC-ClinicDB when using a high-alpha setting ($\alpha = 0.7, \beta = 0.3$), which reached 65.58%. This slight improvement can be attributed to intense specular highlights and mucosal reflections that cause false positives; here, a precision-focused penalty ($\alpha > \beta$) helps restrict mask bleeding. However, this same high-alpha setting drops the accuracy on ISBI-2012 to 52.54%, leading to fragmented membrane predictions. Thus, the asymmetric configuration ($\alpha = 0.3, \beta = 0.7$) was selected as our cross-modality baseline.

Looking at the edge-regularization coefficient λ , the analysis underscores the value of explicit boundary supervision. Omitting the gradient constraint completely ($\lambda = 0.00$) results in lower Jaccard scores of 49.95% on ISBI-2012 and 62.79% on CVC-ClinicDB, showing that pure region-based loss functions are often invariant to exact boundary alignments.

Interestingly, incorporating the boundary loss exhibits non-monotonic trends during these short runs. For example, an initial weight of $\lambda = 0.05$ on CVC-ClinicDB yields a drop to 56.73%, likely due to early gradient conflict between the region and edge terms. Raising the weight to $\lambda = 0.10$ resolves this issue, providing stable scores on both datasets (53.57% on ISBI-2012 and 65.30% on CVC-ClinicDB). Increasing λ beyond this point ($\lambda \geq 0.30$), however, over-constrains the network. The loss forces the model to focus too heavily on high-frequency peripheral noise and specular reflections rather than the overall anatomical structure, degrading performance on both datasets. Based on these observations, the configuration with $\alpha = 0.3, \beta = 0.7$, and $\lambda = 0.10$ was chosen as our optimal baseline configuration.

F. Component-Level Ablation Analysis

To systematically validate the standalone and synergistic contributions of each integrated architectural component within the proposed framework, comprehensive ablation studies were conducted across all four distinct imaging modalities. The quantitative variations in terms of the Jaccard Index (%) and Dice Coefficient (%) are detailed in Table VI and Table VII.

Ablation Evaluation Protocol: To manage the massive computational overhead required to train nine separate topological configurations across multiple imaging targets, a resource-efficient proxy evaluation protocol was strictly implemented for the ablation phase. For the smaller datasets—Fluorescence (Murphy Lab) and Electron Microscopy (ISBI-2012)—the entire image volume was utilized. Conversely, for the significantly larger cohorts—Dermoscopy (ISIC 2018) and Endoscopy (CVC-ClinicDB)—a subset of 500 images was ran-

domly sampled from the primary pool to represent the modality behavior.

Furthermore, instead of executing the exhaustive 150-epoch 5-fold cross-validation loop for every structural variant, the ablation network mutations were trained using a standard isolated 80:20 train-validation partition (holdout split) managed at a controlled compute trajectory. Due to the minimized sample boundaries and single-split evaluation, the absolute baseline metrics reported within this ablation sub-analysis naturally exhibit structural distribution shifts compared to the fully converged, full-dataset cross-validation values reported in Table I and Table III. Nevertheless, this proxy configuration successfully preserves localized trend fidelity, isolating architectural impacts from data-scale effects.

The empirical ablation dynamics reveal highly complex, modality-specific behavior that underscores the necessity of a hybrid architectural design.

In the **Dermoscopy (ISIC 2018)** domain, integrating a pure Multi-Head Self-Attention (MHSA) bottleneck without channel-wise calibration results in an immediate performance decay, with the Jaccard index dropping significantly from 75.41% down to 71.99%. This crucial finding demonstrates that raw Vision Transformers lack inherent local inductive biases, rendering them highly vulnerable to high-frequency surface artifacts such as hair noise and stray skin markings. However, when the SE-Block is concurrently deployed (*Baseline + SE + Transformer*), the Jaccard score stabilizes and recovers back to 73.75%, establishing that adaptive channel-recalibration is mathematically vital to filter out structural noise before global attention projections.

Conversely, the Endoscopy (CVC-ClinicDB) dataset highlights a powerful synergy between channel calibration and global context modeling. Incorporating both SE-Blocks and the MHSA bottleneck simultaneously (*Baseline + SE + Transformer*) achieves a substantial optimization ceiling, yielding the highest Jaccard index of 78.74% and a Dice score of 87.84%. This sharp increase occurs because endoscopic sequences contain irregular polyp morphologies where capturing long-range contextual relationships within a cleaned latent space is critical to preventing spatial under-segmentation.

Furthermore, the structural edge configurations evaluated on the grayscale Electron Microscopy (ISBI-2012) dataset indicate the effectiveness of the boundary-preserving loss strategy. The integrated HTS-MultiResUNet model achieves a Jaccard index of 88.47% and a Dice score of 93.88%. These results suggest that under topological ambiguities where cellular membranes tend to blend into the surrounding background, enforcing structural edge supervision via the discrete derivative Sobel operator can help maintain better geometric contour fidelity.

G. Methodological Limitations and Future Diagnostic Extensions

While the proposed HTS-MultiResUNet demonstrates statistically significant advancements across multiple multimodal imaging benchmarks in terms of region-based metrics (*Dice* and *Jaccard*), a few distinct computational and structural limitations must be explicitly addressed.

TABLE VIII: Component-Level Ablation Study Metrics on Dermoscopy (ISIC 2018) and Endoscopy (CVC-ClinicDB) Datasets

Configuration	SE-Block	MHSA	Att-ResPath	Loss Function	Dermoscopy (ISIC 2018)		Endoscopy (CVC-ClinicDB)	
					Jaccard (%)	Dice (%)	Jaccard (%)	Dice (%)
Baseline MultiResUNet	×	×	×	BCE	75.41	85.79	73.14	83.71
Baseline + Transformer	×	✓	×	BCE	71.99	83.54	75.99	85.85
Baseline + SE	✓	×	×	BCE	75.97	86.07	75.40	85.41
Baseline + Att-ResPath	×	×	✓	BCE	76.18	86.30	71.35	82.66
Baseline + EA-FTL Loss	×	×	×	EA-FTL	76.07	86.18	73.07	83.97
Baseline + SE + Trans	✓	✓	×	BCE	73.75	84.61	78.74	87.84
Baseline + SE + EA-FTL	✓	×	×	EA-FTL	78.52	87.81	75.09	85.14
Baseline + Trans + EA-FTL	×	✓	×	EA-FTL	72.88	84.06	75.56	85.80
HTS-MultiResUNet (Ours)	✓	✓	✓	EA-FTL	75.63	85.67	75.60	85.31

TABLE IX: Component-Level Ablation Study Metrics on Fluorescence (ISBI 2009) and Electron (ISBI-2012) Datasets

Configuration	SE-Block	MHSA	Loss Function	Fluorescence (ISBI 2009)		Electron (ISBI-2012)	
				Jaccard (%)	Dice (%)	Jaccard (%)	Dice (%)
Baseline MultiResUNet	×	×	Static Loss	89.14	94.24	87.46	93.31
Baseline + Transformer	×	✓	Static Loss	88.82	94.06	87.33	93.23
Baseline + SE	✓	×	Static Loss	90.27	94.87	87.84	93.52
Baseline + EA-FTL Loss	×	×	Adaptive Loss	91.22	95.41	88.09	93.67
Baseline + SE + Trans	✓	✓	Static Loss	89.52	94.46	87.67	93.43
Baseline + SE + EA-FTL	✓	×	Adaptive Loss	90.28	94.88	88.24	93.75
Baseline + Trans + EA-FTL	×	✓	Adaptive Loss	91.37	95.48	87.87	93.54
HTS-MultiResUNet (Ours)	✓	✓	Full Loss	91.03	95.29	88.47	93.88

First, regarding geometric boundary assessment, the quantitative evaluation within this study remains primarily bounded within pixel-overlap space. Enforcing edge supervision via the discrete Sobel operator within the Edge-Aware Focal Tversky Loss (*EA-FTL*) successfully guides boundary definition qualitatively, as visualized in our response maps. However, comprehensive surface distance modeling—such as the explicit computation of the Average Symmetric Surface Distance (ASSD) and the 95% Hausdorff Distance (HD95)—requires advanced mesh-grid projections that are currently absent from our metric evaluation pipeline. Consequently, the localized spatial variations along sub-millimeter tissue boundaries cannot be fully decoupled from global pixel errors in this iteration.

Second, although the spatial-channel gating inside the *Att-ResPath* pathways actively operates as a feature filter to purge high-frequency peripheral clutter (effectively reducing False Positive triggers caused by specular highlights or skin hairs), a deterministic trade-off regarding the False Negative (FN) profile remains unquantified. From a clinical perspective, reducing false negatives is particularly important because missed lesions may affect diagnostic accuracy. The current macro-topographical loss weights freeze the classification threshold uniformly at 0.5, which may induce over-conservative contour contractions on micro-scale sessile polyps.

Finally, from a computational efficiency perspective, integrating long-range Multi-Head Self-Attention (MHSA) token projections inside the latent bottleneck inevitably introduces an operational memory footprint overhead. Although limiting the MHSA deployment strictly to the deepest bottleneck layer prevents out-of-memory errors on limited-resource worksta-

tions (such as standard NVIDIA Tesla P100 setups); however, inference latency and frame-per-second (FPS) throughput were not quantitatively evaluated. Consequently, the computational suitability of the proposed architecture for latency-sensitive applications remains to be investigated.

Addressing these computational trade-offs through low-precision hardware quantization, evaluating explicit surface distance metrics, and optimizing asymmetric hyperparameter tuning to strategically minimize False Negatives represent important foundations for our subsequent investigations.

VIII. DISCUSSION AND CONCLUSION

A. Research Limitations and Open Challenges

Despite the consistent performance trends established across the multi-modal biomedical benchmarks, several structural and operational limitations remain that offer avenues for future investigation. Architecturally, the current configuration of HTS-MultiResUNet is strictly constrained to the processing of 2D planar image slices. While effective for surface dermoscopic scans and endoscopic video frame extractions, this topological layout inherently disregards the continuous cross-slice anatomical contexts available within dense 3D clinical volumes, such as computed tomography (CT) or magnetic resonance imaging (MRI) sequences.

Furthermore, although the strategic placement of the Multi-Head Self-Attention (MHSA) bottleneck at the deepest layer keeps the aggregate footprint within a controlled budget of 11.08 million parameters, the dense multi-scale convolutional factorizations inside the MultiRes blocks still pose a non-trivial computational cost during the dense spatial gradient

training phase. This complexity restricts the architecture from being natively trained on extreme high-resolution whole-slide imaging (WSI) frameworks without extensive sub-patching strategies. Finally, while our parametric sensitivity sweep successfully identified stable numerical regions for the loss coefficients ($\alpha = 0.3, \beta = 0.7, \lambda = 0.10$), the framework still relies on empirical offline grid-search schedules, highlighting the need for dynamic, data-driven hyperparameter modulation frameworks.

B. Concluding Summary and Core Contributions

In this paper, we introduced HTS-MultiResUNet, a hybrid deep learning architecture designed to address the receptive field constraints of conventional convolutional networks in multimodal biomedical image segmentation tasks. By integrating Squeeze-and-Excitation (SE) mechanisms and a localized Hybrid Transformer Bottleneck into the MultiResUNet baseline framework, the proposed model balances localized hierarchical feature extraction with abstract, long-range global contextual reasoning.

Furthermore, the introduction of the Attention-aware Residual Path (Att – ResPath) reduces the semantic mismatch gap between matching encoder and decoder resolution stages by filtering out raw background details. Concurrently, the network is optimized via the formulated Edge-Aware Focal Tversky Loss (EA – FTL) function, utilizing a discrete derivative Sobel operator to enforce structural boundary constraints while handling spatial class imbalances.

The performance of the proposed framework was evaluated using a five-fold cross-validation protocol across four medical imaging modalities: Fluorescence Microscopy, Electron Microscopy, Dermoscopy, and Endoscopy. Compared to the baselines, the integrated model yields marginal score improvements. Results from the ablation study indicate that while individual attention variants show localized limitations in noisy regimes—such as dermoscopic hair artifacts—the unified model offers a modest positive contribution to the overall segmentation output.

Quantitatively, the proposed architecture demonstrated consistent validation advantages across both the Jaccard Index and the Dice Coefficient metrics, backed by a paired Wilcoxon signed-rank test confirming statistical significance ($p < 0.043$). Most notably, a relative improvement of 5.83% in Jaccard score was registered within the artifact-laden Dermoscopy dataset, pushing the performance metrics to $72.60\% \pm 2.20\%$. Qualitatively, visual segmentation masks confirmed that HTS-MultiResUNet exhibits enhanced resistance against complex clinical perturbations—such as blinding specular highlights, mucosa textures, and marker ink trails—suppressing disconnected false-positive artifacts while preserving the geometric fidelity of intricate morphological boundaries.

C. Future Research Directions

Consequently, future research pathways will focus on extending the HTS-MultiResUNet formulations into a fully volumetric 3D network architecture. This structural scaling will involve upgrading the internal multi-scale extraction paths,

Att-ResPath routing highways, and self-attention operations into a 3D tensor space to process spatial continuity across independent volumetric slices.

Additionally, we plan to explore advanced structural model compression techniques—including structured network pruning, deep layer-wise quantization, and structural reparameterization pipelines. These optimization streams are intended to reduce the parametric footprint of the hybrid architecture, accelerating inference speeds and facilitating future deployment on resource-constrained computing platforms.

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